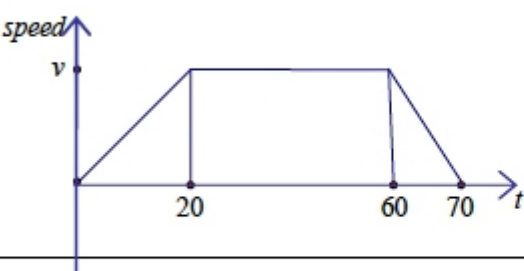
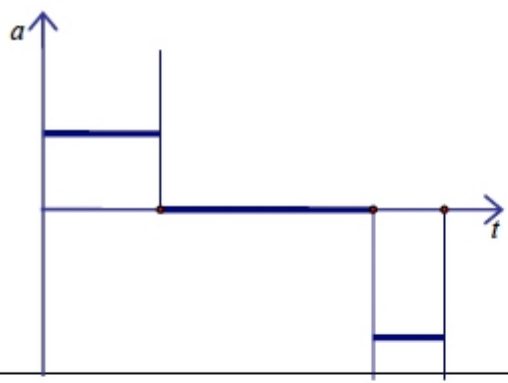


A level Applied Paper 3B Mechanics Practice Paper J12 **MARK SCHEME**

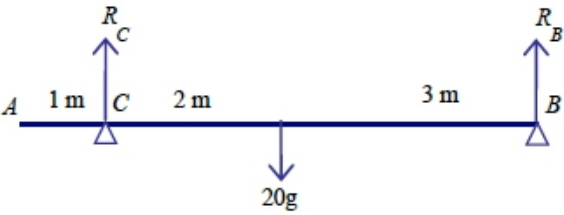
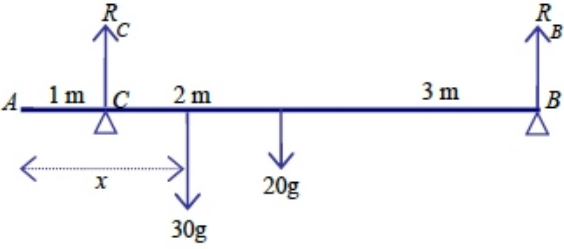
Question 1

Question Number	Scheme	Marks
(a) (i)	 1st section correct 2nd & 3rd sections correct Numbers and v marked correctly on the axes.	B1 B1 DB1
(ii)	 1st section correct 2nd section correct 3rd section correct and no "extras" on the sketch	B1 B1 B1 (6)
(b)	$\frac{70 + 40}{2} \times v = 880$ $v = 880 \times \frac{2}{110} = 16$	M1 A1 DM1 A1 (4) [10]

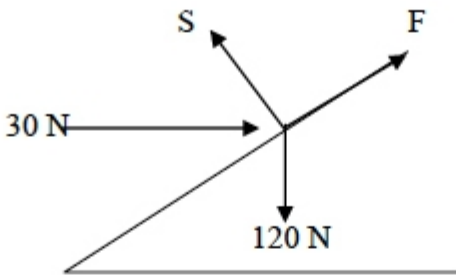
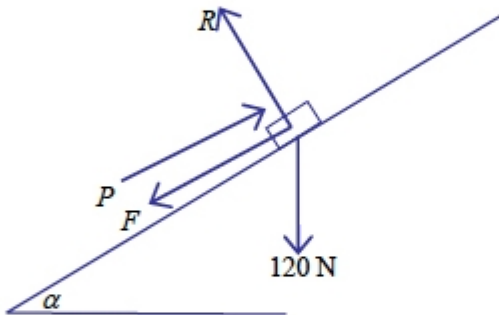
Question 2

Question Number	Scheme	Marks
(a)	$-6.45 = u - 9.8 \times 0.75$ $0.9 = u$ **	M1 A1 A1 (3)
(b)	$0 = 0.81 - 2 \times 9.8 \times s$ $s = 0.041$ or 0.0413	M1 A1 (2)
(c)	$h = -0.9 \times 0.75 + 4.9 \times 0.75^2$ $h = 2.1$ or 2.08	M1 A1 A1 (3) [8]

Question 3

Question Number	Scheme	Marks
(a)	 Taking moments about B: $5 \times R_C = 20g \times 3$ $R_C = 12g$ or $60g/5$ or 118 or 120 Resolving vertically: $R_C + R_B = 20g$ $R_B = 8g$ or 78.4 or 78	M1A1 A1 M1 A1 (5)
(b)	 Resolving vertically: $50g = R + R$ Taking moments about B: $5 \times 25g = 3 \times 20g + (6 - x) \times 30g$ $30x = 115$ $x = 3.8$ or better or $23/6$ oe	B1 M1 A1 A1 A1 (5) [10]

Question 4

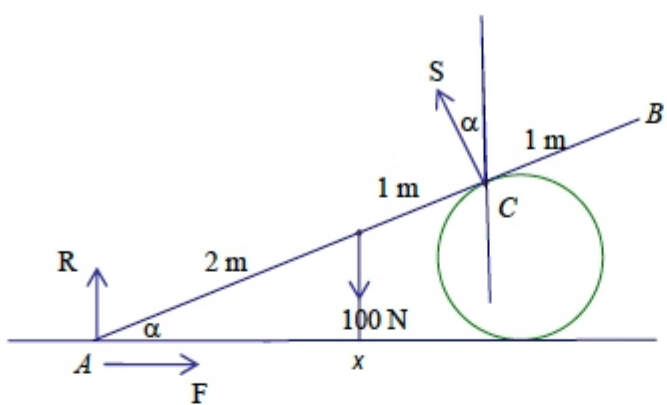
Question Number	Scheme	Marks
(a)	 Resolving perpendicular to the plane: $S = 120 \cos \alpha + 30 \sin \alpha$ $= 114 *$	M1 A1 A1 A1 (4)
(b)	 Resolving perpendicular to the plane: $R = 120 \cos \alpha$ $= 96$ $F_{\max} = \frac{1}{2} R$ Resolving parallel to the plane: In equilibrium: $P_{\max} = F_{\max} + 120 \sin \alpha$ $= 48 + 72 = 120$	M1 A1 A1 M1 M1 A(2,1,0) A1 (8)
(c)	$30 + F = 120 \sin \alpha$ OR $30 - F = 120 \sin \alpha$ So $F = 42\text{N}$ acting up the plane.	M1 A1 A1 (3) [15]



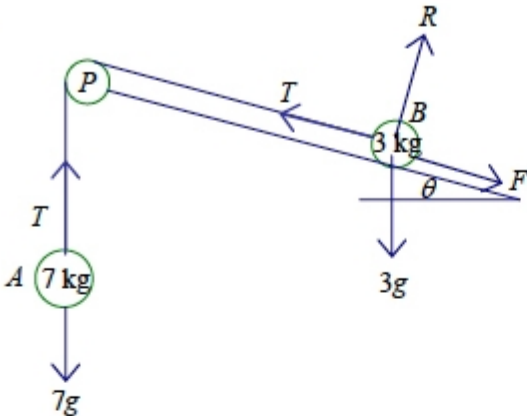
Question 5

Question Number	Scheme	Marks
(a)	$\text{speed} = \sqrt{2^2 + (-5)^2}$ $= \sqrt{29} = 5.4 \text{ or better}$	M1 A1 (2)
(b)	$((7\mathbf{i} + 10\mathbf{j}) - (2\mathbf{i} - 5\mathbf{j}))/5$ $= (5\mathbf{i} + 15\mathbf{j})/5 = \mathbf{i} + 3\mathbf{j}$ $\mathbf{F} = m\mathbf{a} = 2(\mathbf{i} + 3\mathbf{j}) = 2\mathbf{i} + 6\mathbf{j}$	M1 A1 A1 DM1 A1ft (5)
(c)	$\mathbf{v} = \mathbf{u} + \mathbf{a}t = (2\mathbf{i} - 5\mathbf{j}) + (\mathbf{i} + 3\mathbf{j})t$ $(-5 + 3t)\mathbf{j}$ Parallel to $\mathbf{i} \Rightarrow -5 + 3t = 0$ $t = 5/3$	M1 A1 M1 A1 (4) [11]

Question 6

Question Number	Scheme	Marks
	 Taking moments about A: $3S = 100 \times 2 \times \cos \alpha$ Resolving vertically: $R + S \cos \alpha = 100$ Resolving horizontally: $S \sin \alpha = F$ (Most alternative methods need 3 independent equations, each one worth M1A1. Can be done in 2 e.g. if they resolve horizontally and take moments about X then $R \times 2 \times \cos \alpha = S \times (3 - 2 \times \cos^2 \alpha)$ scores M2A2) Substitute trig values to obtain correct values for F and R (exact or decimal equivalent). $\left(S = \frac{200\sqrt{8}}{9} \right), R = 100 - \frac{1600}{27} = \frac{1100}{27} \approx 40.74, F = \frac{200\sqrt{8}}{27} \approx 20.95...$ $F \leq \mu R, 200\sqrt{8} \leq \mu \times 1100, \mu \geq \frac{200\sqrt{8}}{1100} = \frac{2\sqrt{8}}{11}.$ Least possible μ is 0.514 (3sf), or exact.	M1 A1 M1 A1 M1 A1 DM1 A1 M1 A1 [10]

Question 7

Question Number	Scheme	Marks
(a)	 $\tan \theta = \frac{5}{12}$ $\sin \theta = \frac{5}{13}$ $\cos \theta = \frac{12}{13}$ For A: $7g - T = 7a$ For B: parallel to plane $T - F - 3g \sin \theta = 3a$ perpendicular to plane $R = 3g \cos \theta$ $F = \mu R = 3g \cos \theta = 2g \cos \theta$ Eliminating T, $7g - F - 3g \sin \theta = 10a$ Equation in g and a: $7g - 2g \times \frac{12}{13} - 3g \times \frac{5}{13} = 7g - \frac{39}{13}g = 4g = 10a$ $a = \frac{2g}{5}$ oe or 3.9 or 3.92	M1 A1 M1 A1 M1 A1 M1 DM1 DM1 A1 (10)
(b)	After 1 m, $v^2 = u^2 + 2as$, $v^2 = 0 + 2 \times \frac{2g}{5} \times 1$ $v = 2.8$	M1 A1 (2)
(c)	$-(F + 3g \sin \theta) = 3a$ $\frac{2}{3} \times 3g \times \frac{12}{13} + 3g \times \frac{5}{13} = 3g = -3a$, $a = -g$ $v = u + at$, $0 = 2.8 - 9.8t$, $t = \frac{2}{9.8}$ oe, 0.29, 0.286	M1 A1 DM1 A1 (4) [16]



Question 8

(a)	$a = 4t^3 - 12t$ Convincing attempt to integrate $v = t^4 - 6t^2 (+c)$ Use initial condition to get $v = t^4 - 6t^2 + 8 (\text{ms}^{-1})$.	M1 A1 A1 (3)
(b)	Convincing attempt to integrate $s = \frac{t^5}{5} - 2t^3 + 8t (+0)$ Integral of their v	M1 A1ft (2)
(c)	Set their $v = 0$ Solve a quadratic in t^2 $(t^2 - 2)(t^2 - 4) = 0 \Rightarrow$ at rest when $t = \sqrt{2}, t = 2$	M1 DM1 A1 (3) [8]



Question 9

Question Number	Scheme	Marks
9		
(a)	i → distance = $6t$ j ↑ distance = $12t - \frac{1}{2}gt^2$ At B, $2\left(12t - \frac{1}{2}gt^2\right) = 6t$ $(24 - 6)t = gt^2$ $18 = gt, t = \frac{18}{g} (= 1.84s)$	B1 M1 A1 M1 A1 M1 A1 (7)
(b)	i → speed = 6 j ↑ velocity = $12 - gt = -6$ ∴ speed at A $= \sqrt{6^2 + 6^2} = \sqrt{72} = 6\sqrt{2} (= 8.49)(ms^{-1})$	B1 M1 A1 M1 A1 (5)
(c)	↑ speed = $12 - gt = +6$ $t = \frac{6}{g} (= 0.61s)$	M1 A1 ft A1 (3) (15 marks)